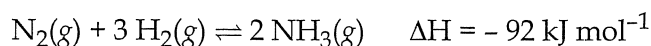


1. [6 marks]

2.2.4, 2.2.7 (2015:30)

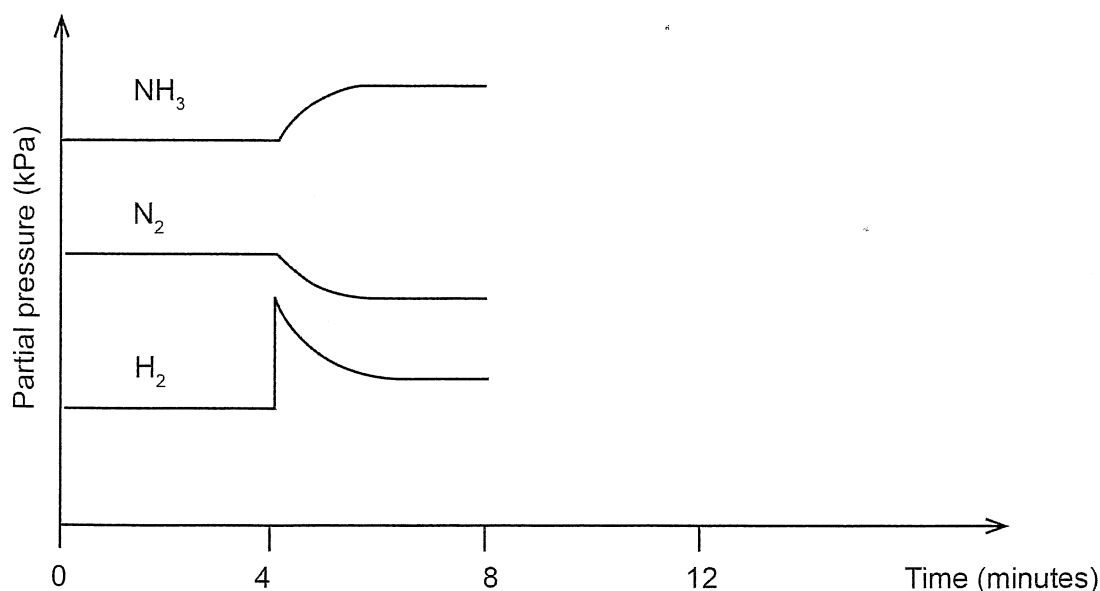
Ammonia exists in equilibrium with hydrogen and nitrogen as shown by the following exothermic equation.



As they exist in the gaseous state, the relative concentrations can be given in terms of the partial pressure (kPa) of each gas.

Nitrogen, hydrogen and ammonia gases are placed in a rigid container and allowed to reach equilibrium. The graph below shows the partial pressures of the gaseous system initially at equilibrium. After the experiment operates for 4 minutes, a change is imposed upon it.

Partial pressures of NH_3 , N_2 and H_2 over time



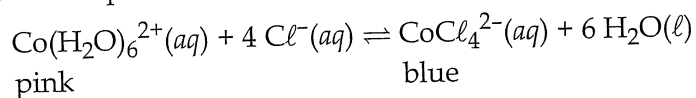
- (a) What characteristic of equilibrium is indicated on the graph by the section from 0 to 4 minutes? [1]

- (b) A change was imposed on the system at the 4 minute mark. What imposed change could have produced the results indicated on the graph? [1]

- (c) The system was **suddenly** cooled at 8 minutes and then reached equilibrium again at 12 minutes. Using this information, complete the graph above from the 8 to the 12 minute mark. [4]

2. [16 marks]

The two different coloured cobalt(II) complex ions, $\text{Co}(\text{H}_2\text{O})_6^{2+}$ and CoCl_4^{2-} , exist together in equilibrium in solution in the presence of chloride ions. This is represented by the equation below.



An experiment is conducted to investigate the effects on the equilibrium position by imposing a series of changes on the system. The shift in equilibrium position can be indicated by any colour change of the solution.

Colour chart	
Species	Colour
$\text{Co}(\text{H}_2\text{O})_6^{2+}(\text{aq})$	pink
$\text{CoCl}_4^{2-}(\text{aq})$	blue
Initial equilibrium mixture	purple

After a 3.00 mL sample of an initial equilibrium mixture was placed in each of three test tubes, changes to each system were made by adding a different substance, as indicated in the table below.

Test tube	Substance added to the test tube
1	10 to 12 drops of distilled water
2	20 to 25 drops of concentrated hydrochloric acid
3	20 to 25 drops of 0.200 mol L ⁻¹ silver nitrate solution, $\text{AgNO}_3(\text{aq})$

- (a) Complete the table below by predicting the:
- change in concentration, if any, of each of the ions in solution compared to the initial solution, after a new equilibrium position is reached.
 - colour change, if any, that takes place from the initial purple-coloured solution. [6]

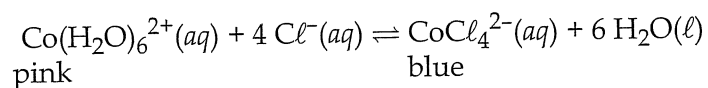
Additions to the test tube	Change in concentration from initial equilibrium to final equilibrium (increase, decrease, unchanged)			Colour favoured (pink, blue or unchanged)
	$[\text{Co}(\text{H}_2\text{O})_6^{2+}]$	$[\text{Cl}^-]$	$[\text{CoCl}_4^{2-}]$	
1. add $\text{H}_2\text{O}(\ell)$				
2. add $\text{HCl}(\text{aq})$				
3. add $\text{AgNO}_3(\text{aq})$				

- (b) Other than a colour change, what else should be observed in test tube 3? [1]

- (c) Using Collision Theory, explain your predicted observations when hydrochloric acid is added to test tube 2. [3]

Another experiment was conducted to investigate the effect that changing the temperature had on the equilibrium mixture. When 3.00 mL of the original equilibrium mixture was placed in a test tube and then in an ice bath, the solution became pink.

- (d) Determine whether the forward reaction, as illustrated by the equation below, is exothermic or endothermic. Use Le Châtelier's Principle to justify your answer. [4]



- (e) State **one** specific hazard to the environment that the disposal of chemicals from this experiment poses and state what could be done in the laboratory to reduce this hazard. [2]

3. [6 marks]

2.7 (2016 SP:34)

Ocean acidification results from carbon dioxide dissolving in water and an equilibrium being established between the water and carbon dioxide to produce carbonic acid, (H_2CO_3).

- (a) Write a balanced equation for this equilibrium. [2]

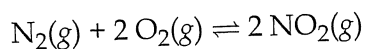
- (b) The formation of carbonic acid leads to an increase in the hydronium ion (H_3O^+) concentration in water. Show the equilibrium that results in the formation of hydronium ions when carbonic acid reacts with water. [1]

- (c) State **one** problem ocean acidification is causing for marine organisms. Explain how this problem arises and support your answer with an appropriate balanced equation. [3]

2.2.11, 1.4 (2016:41)

4. [18 marks]

Nitrogen dioxide is toxic to humans when inhaled and is a significant component of air pollution. It can be formed by the combustion of nitrogen in the air at high temperatures; firstly forming nitric oxide NO(g) and on further oxidation, forming nitrogen dioxide, $\text{NO}_2\text{(g)}$. The overall equation for this process is given here:

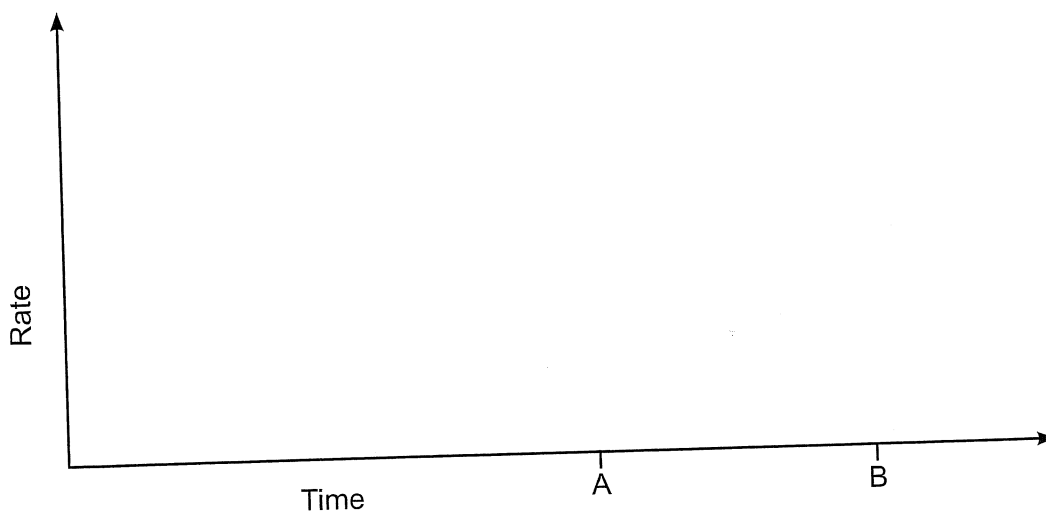


The following questions relate to the equilibrium system represented by this equation.

- (a) Write the equilibrium expression for this reaction when it is in equilibrium. [2]

- (b) Assuming all other conditions remain constant, what happens to the equilibrium constant after the pressure of the system is lowered and equilibrium is re-established? [1]

- (c) (i) On the axes below, draw the forward (—) and reverse (---) reaction rates, starting at the moment the oxygen and nitrogen gases begin to react with each other until after equilibrium has been established at time A. Continue the graph until time B. [3]

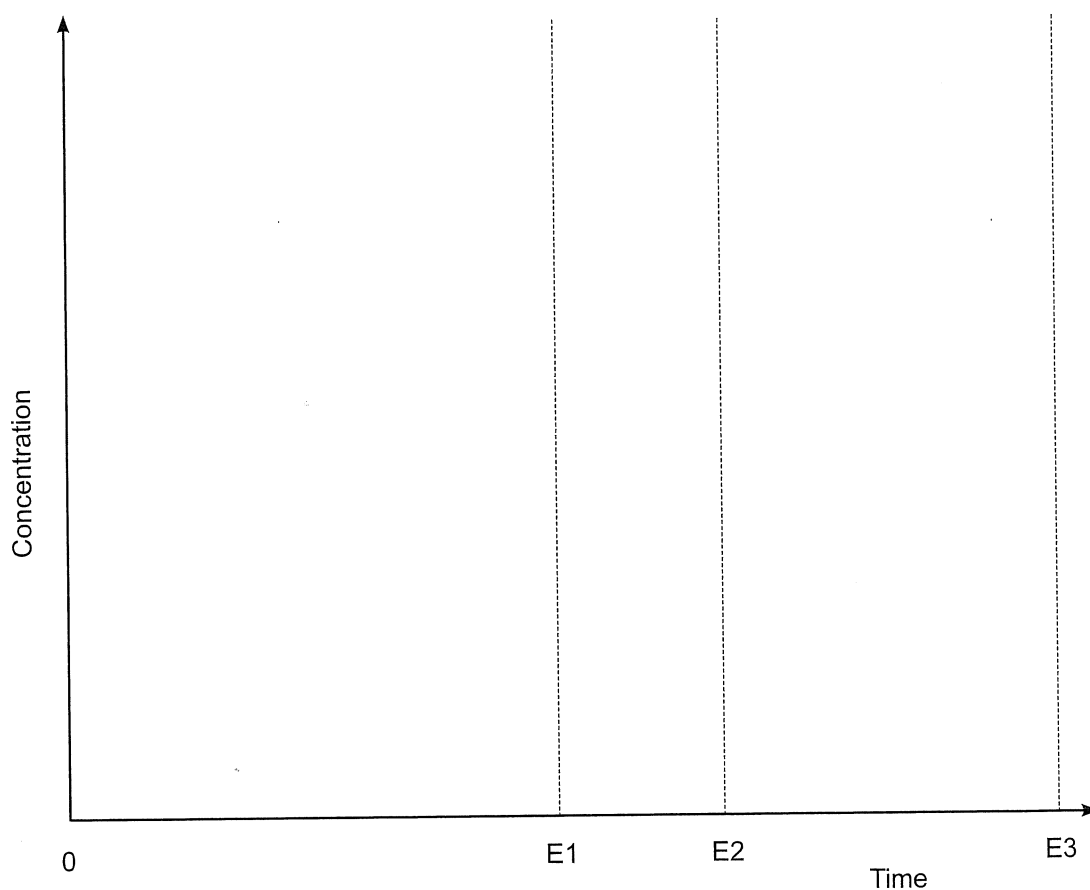


- (ii) On the same axes above, draw and label clearly the effect of conducting the same reaction at a higher temperature. [2]

- (d) On the axes below, draw separate curves to show how the concentrations of the **three** gases change with time, starting at the moment the oxygen and nitrogen gases begin to react with each other until the system reaches equilibrium at Time E1. Continue the graph from Time E1 to Time E2. Assume that the initial concentrations of oxygen and nitrogen are identical.

Label clearly the line for each gas.

[5]



- (e) At Time E2 shown on the axis, the reaction vessel is doubled in volume, and the system is then again allowed to reach equilibrium at Time E3. On the same graph above, show how the concentrations of the three gases would change in response to the change in volume, from Time E2 until equilibrium is re-established at Time E3.

[3]

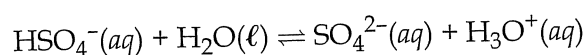
The reaction between nitrogen gas and oxygen gas occurs at high temperatures such as those found in the combustion engines of cars. The atmosphere is composed of 78% nitrogen and 21% oxygen and has been stable for millions of years.

- (f) What does the stability of this composition indicate about the equilibrium constant and energy requirements of the reaction between nitrogen and oxygen gases?

[2]

5. [10 marks]

A hydrogensulfate/sulfate system is represented by the following equation.



(a) Predict how

- the forward reaction rate and
- the pH

will differ from their original values after the following changes are imposed on the system and equilibrium has been re-established. Use the terms increase, decrease, no change. [6]

Change imposed by the addition of	Effect on forward reaction rate when equilibrium is re-established	Effect on pH when equilibrium is re-established
a few drops of concentrated hydrochloric acid		
a few drops of concentrated lead(II) nitrate solution		
distilled water		

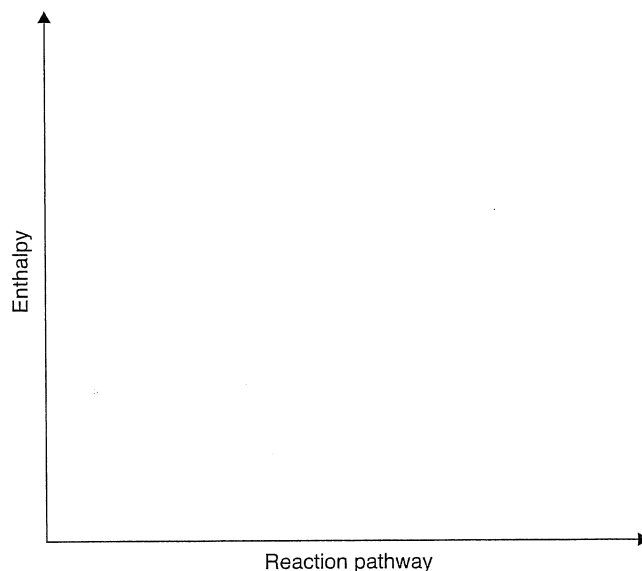
(b) The reaction in part (a) is endothermic in the forward direction as written. Predict what will happen to the pH when the temperature is increased. Justify this prediction. [4]

6. [9 marks]

2.2.7 (2017:33)

Both dynamite and TNT are explosive substances that are sometimes confused with each other. The active ingredient in dynamite is a stabilised form of nitroglycerine, $\text{C}_3\text{H}_5\text{N}_3\text{O}_9$, while TNT is the common name for the explosive compound 2,4,6-trinitrotoluene, $\text{C}_6\text{H}_2(\text{NO}_2)_3\text{CH}_3$.

- (a) An explosion is a 'very fast and very exothermic reaction'. Use a solid line (—) to draw, and then label, an energy profile diagram reflecting an explosive reaction. [3]



Nitroglycerin is extremely shock-sensitive and readily becomes unstable. In dynamite the nitroglycerin is combined with inhibitors and stabilisers, making it safer to use. Typically, dynamite is between 25% to 50% nitroglycerin.

- (b) An inhibitor is a substance that decreases the rate of, or prevents, a chemical reaction. On the diagram in part (a) above, indicate by way of a dashed line (----) any change/s that would be evident if an inhibitor were to be introduced. [2]
- (c) The energy density of dynamite is 5.0 MJ kg^{-1} and the energy density of TNT is 4.0 MJ kg^{-1} . Show by calculation and by reasoning which of these two explosives produces more energy per mole of the active ingredient. [4]

$$M(\text{C}_3\text{H}_5\text{N}_3\text{O}_9) = 227.10 \text{ g mol}^{-1}$$

$$M(\text{C}_6\text{H}_2(\text{NO}_2)_3\text{CH}_3) = 227.14 \text{ g mol}^{-1}$$

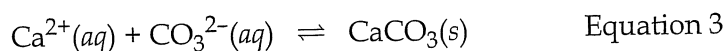
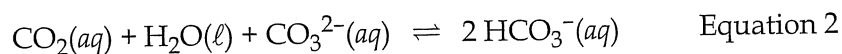
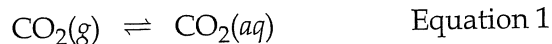
2.1 (2018:32)

7. [4 marks]

7. Many marine animals have shells that consist mainly of calcium carbonate. These shells are built from dissolved calcium and carbonate ions.

As the amount of atmospheric carbon dioxide increases, more carbon dioxide dissolves in the ocean. There is increasing concern that as more carbon dioxide dissolves, it will be more difficult for calcium carbonate to form.

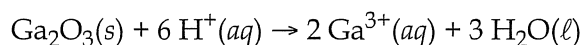
Use the following equations to explain why an increasing concentration of atmospheric carbon dioxide will decrease the formation of calcium carbonate

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There is no handwriting or other markings on the paper.

8. [12 marks]

2.2.1, 1.4 (2018:37)

Gallium is present as gallium(III) oxide, Ga_2O_3 , in the red mud waste from the processing of bauxite. The first step in its recovery from the red mud is the addition of hydrochloric acid, $\text{HCl}(\text{aq})$. This is represented by the equation below.

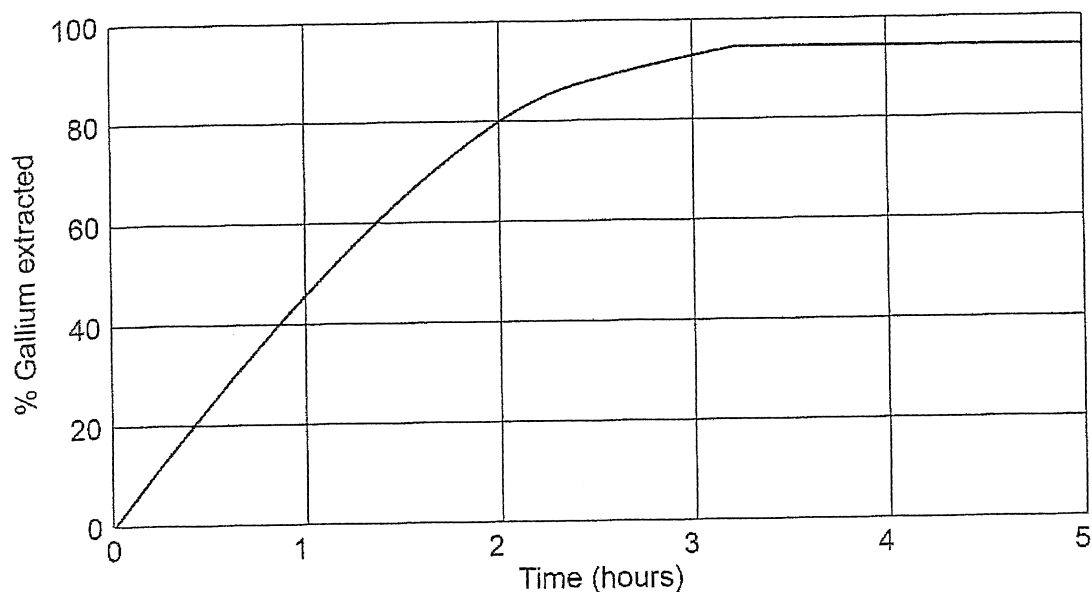


The results in the table below show the effect of temperature on the rate of gallium extraction from a red mud sample. Note that all of the other reaction conditions were constant.

Temperature (°C)	Reaction rate (as percentage of gallium extracted after four hours)
40	77
55	88
70	95
85	96
100	97

- (a) Explain the effect of temperature on reaction rate by applying collision theory. Support your explanation with an appropriate and clearly-labelled diagram. [7]

This graph shows how the amount of gallium extracted from red mud varies over time at a hydrochloric acid concentration of 1.00 mol L^{-1} .



- (b) Sketch on the graph above the result that would be obtained if the hydrochloric acid concentration was changed to 2.00 mol L^{-1} . [2]
- (c) Use collision theory to justify the position and shape of the graph you sketched in part (b). Assume that all other reaction conditions were kept constant. [2]

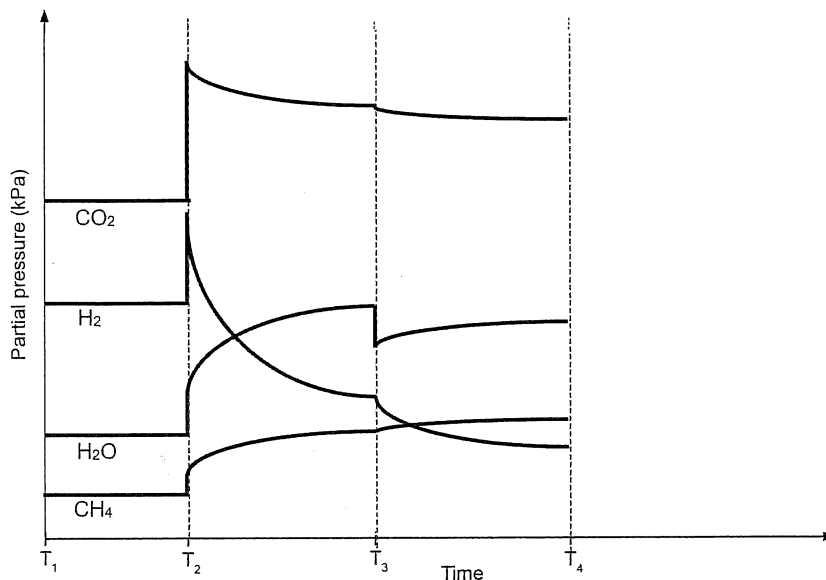
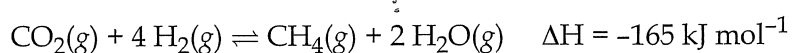
In a laboratory analysis, the red mud containing gallium(III) oxide, was mixed with excess hydrochloric acid solution. The concentration of gallium(III) ions, $\text{Ga}^{3+}(\text{aq})$, in the resulting solution was analysed and the percentage of gallium in the red mud was determined.

- (d) State **one** reason why the hydrochloric acid used in this analysis needed to be in excess. [1]

- (b) Predict the effect of each of the following changes on the methane yield in the Sabatier reaction. [5]

Imposed change	Effect on methane yield (circle your answer)		
a suitable catalyst is added	increase	decrease	no effect
the volume of the reaction vessel is increased	increase	decrease	no effect
the temperature is increased	increase	decrease	no effect
methane is removed through a special valve as soon as it forms	increase	decrease	no effect
the partial pressure of carbon dioxide is decreased	increase	decrease	no effect

Graphs can be drawn to show the effects of imposed changes on equilibrium systems. The graph below shows the effects of some changes that might be made to the reacting system in a flexible vessel.



With reference to the above graph, answer the following questions.

(c) (i) What happened at T_1 ? [1]

(ii) Identify the change imposed at each time in the table below. [2]

Time	Change imposed on the system
T_2	
T_3	

(iii) The temperature of the reaction vessel was decreased at T_4 . Sketch on the graph above to show how this affected the partial pressures of all species present. Include any changes to scale and continue until a new equilibrium is established. [3]

10. [13 marks]

3.2.4, 2.2.7 (2019:27)

Calcium hypochlorite, $\text{Ca}(\text{OCl})_2(\text{s})$, is used for the treatment of water in swimming pools and is sold as 'pool chlorine'.

- (a) Explain why a basic solution is produced when 'pool chlorine' is dissolved in the pool water. Include an equation in your answer. [4]

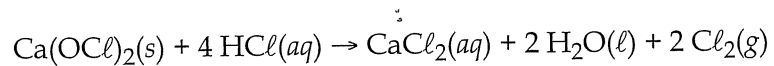
Equation

A pool chemical used to counteract the basicity of the pool water is hydrochloric acid, $\text{HCl}(\text{aq})$. It is sold as 'pool acid'.

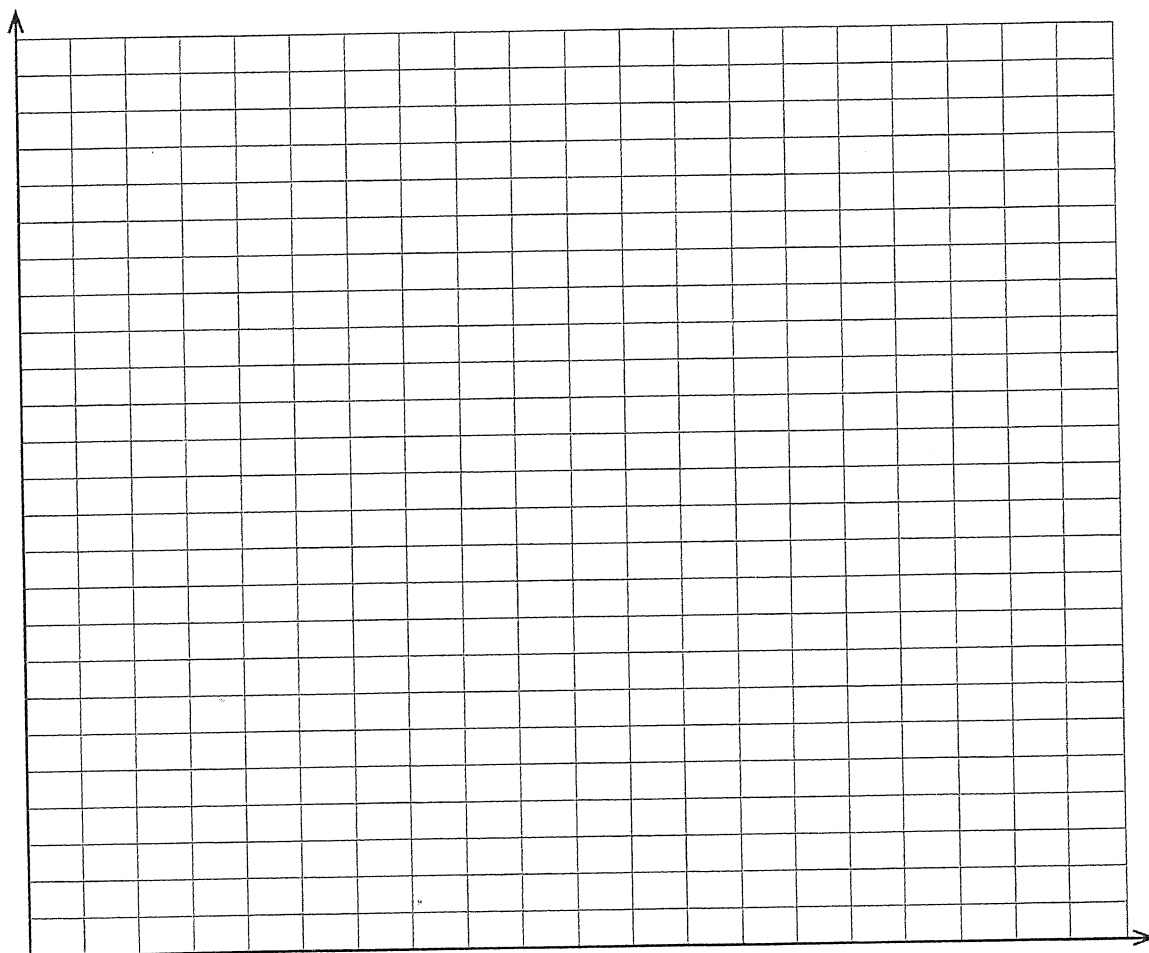
- (b) State what happens to the pH of the pool water when 'pool acid' is added to the pool water. Include an equation to illustrate your statement. [3]

Equation

'Pool chlorine' and 'pool acid' must be stored separately from each other because calcium hypochlorite can react explosively on contact with hydrochloric acid. The equation for this reaction is given below.



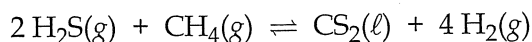
- (c) Sketch a clearly-labelled energy profile diagram illustrating the reaction between the 'pool chlorine' and the 'pool acid'. [6]



11. [11 marks]

2.2.9, 2.2.10 (2020:29)

Some hydrogen sulfide and methane were sealed inside a reaction vessel and the following equilibrium was established:



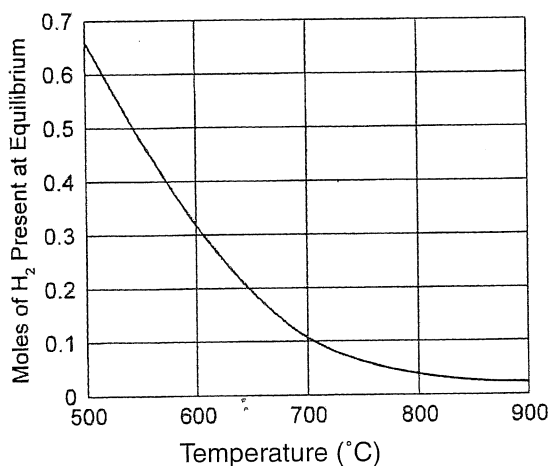
- (a) Write the equilibrium constant expression (K) for this reaction system.

[2]

- (b) Some methane was removed from the reaction vessel. What effect did this have on the position of the equilibrium? Use collision theory to justify your answer.

[5]

The temperature inside the reaction vessel was increased. The heating process was stopped every so often and, once equilibrium had been established at the attained temperature, the amount of hydrogen present in the system was measured. The results are shown on the following graph.



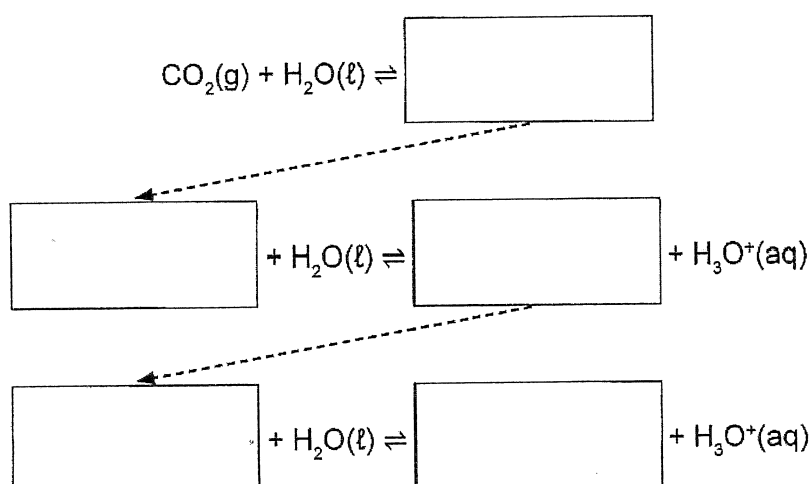
- (c) Using the graph and your answer to part (a), predict the effect of an increase in temperature on the numerical value of K . Justify your prediction. [4]

12. [9 marks]

2.1 (2020:31)

The amount of carbon dioxide in the Earth's atmosphere is increasing, leading to more carbon dioxide dissolving in the oceans and hence ocean acidification.

- (a) Complete the following sequence of equations to show what happens to carbon dioxide when it dissolves in water. [3]



- (b) Other than death, state **two** consequences of the above sequence of equations on marine organisms with shells. [2]

One: _____

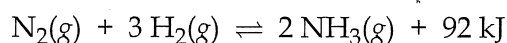
Two: _____

- (c) Use Le Châtelier's Principle and the sequence of equations in part (a) to predict what might happen, in relation to ocean acidification, if the United Nations Kyoto Protocol is discarded. Explain your reasoning. [4]

13. [10 marks]

2.2.9 (2020:34)

The Haber process is used to make ammonia. The balanced equation for the process is



The Haber process provides challenges for industrial operators in relation to the rate of ammonia production and the ammonia yield. This is reflected in the following quotation taken from *Chemistry and Engineering News*:

'The reaction has good yields at very low temperatures ... but the rate is sluggish. To speed it up, chemists raise the temperature. But at those high temperatures, the reaction's thermodynamics change, and the yield goes down.'

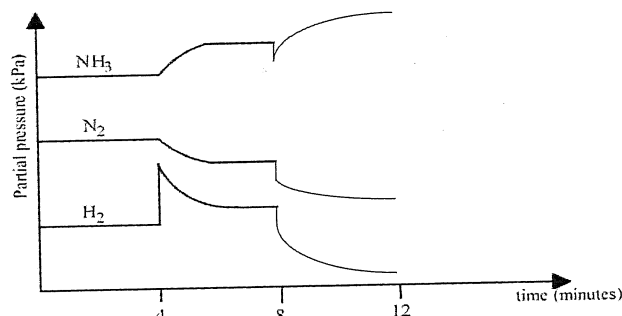
- (a) State whether you agree with the claims about the effects of temperature on the yield of ammonia. Justify your statement using Le Châtelier's Principle. [4]

- (b) State whether you agree with the claims about the effects of temperature on the rate of the Haber process. Justify your statement using collision theory. [6]

- 7.(2018:12-c) A burette can be read to two decimal places and, unlike most glassware, zero is at the top. The error is half a graduation (0.05 mL). Errors are added when data is added or subtracted. So the error is 0.10 mL. The answer is 'c'.
- 8.(2018:13-d) Result 1 appears to be an outlier. It differs from the rest by 0.7 mL – a large amount. Thus, the remaining five values are averaged. The answer is 19.96 mL. The answer is 'd'.
- 9.(2018:16-c) Ethics have not previously been in a SCSA ATAR Chemistry examination. Items (i) and (v) are related to experimental design. The remainder concern doing research fairly and honestly. These relate to ethics. The answer is 'c'.
- 10.(2019:19-a) A hypothesis must relate to empirical data relating a dependent variable and an independent variable and be testable. The only answer to meet these criteria is 'a'.
- 11.(2019:20-d) This question involves understanding of the syllabus terms 'reliability' and 'validity'. Does the test measure what it is supposed to measure and are the results consistent? The answer is 'd'.
- 12.(2020:25) For the burette and pipette the 'solution to be used' is the final rinse and the conical flask is rinsed with distilled water. The answer is (b).

Chapter 7: Chemical Equilibrium

- 1.(2015:30) a) Between 0 and 4 minutes, the partial pressures of all gases were constant.
b) H_2 gas was added.
c) There was an immediate decrease, then the $[\text{NH}_3]$ rose and the others fell in stoichiometric ratios, then equilibrium was established.



- 2.(2015:38) a)

Additions to the test tube	Change in concentration from initial equilibrium to final equilibrium (increase, decrease, unchanged)			Colour favoured (pink, blue or unchanged)
	$[\text{Co}(\text{H}_2\text{O})_6^{2+}]$	$[\text{Cl}^-]$	$[\text{CoCl}_4^{2-}]$	
1. add $\text{H}_2\text{O}(\ell)$	decrease	decrease	decrease	pink
2. add $\text{HCl}(\text{aq})$	decrease	increase	increase	blue
3. add $\text{AgNO}_3(\text{aq})$	increase	decrease	decrease	pink

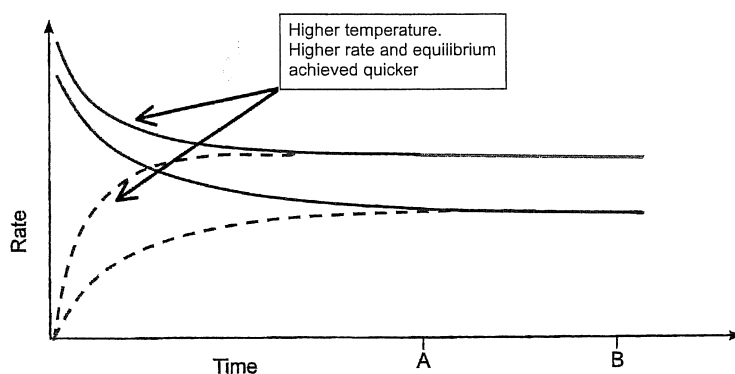
- b) A white precipitate forms
- c) Adding $\text{HCl}(\text{aq})$ increases the concentration of chloride ions and the number of collisions between $\text{Co}(\text{H}_2\text{O})_6^{2+}$ and Cl^- increases. The rate of the forward reaction increases relative to the reverse reaction and hence equilibrium shifts to the right. This leads to a greater concentration of the blue CoCl_4^{2-} and a lower concentration of the pink $\text{Co}(\text{H}_2\text{O})_6^{2+}$ ion, so the solution looks more blue.
- d) Decreasing the temperature shifts the equilibrium position to the left favouring the production of $\text{Co}(\text{H}_2\text{O})_6^{2+}(\text{aq})$, a pink solution. According to Le Châtelier's Principle decreasing the temperature favours the exothermic reaction to oppose the change. Since the reverse reaction has been favoured, that is the exothermic reaction. So the forward reaction is endothermic.
- e) Any hazard relevant to this specific experiment. eg the disposal of concentrated hydrochloric acid. Any method that is relevant eg dilute with large amounts of water when emptying down the sink.

- 3.(2016 SP:34) a) $\text{CO}_2(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{H}_2\text{CO}_3(\text{aq})$
b) $\text{H}_2\text{CO}_3(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{HCO}_3^-(\text{aq})$
c) $\text{H}_3\text{O}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) \rightleftharpoons \text{HCO}_3^-(\text{aq}) + \text{H}_2\text{O}(\ell)$
By lowering the pH more $\text{H}^+(\text{aq})$ is available to convert carbonate ion to hydrogencarbonate, lowering the $[\text{CO}_3^{2-}]$ and so limiting the ability of coral to form calcium carbonate reef.

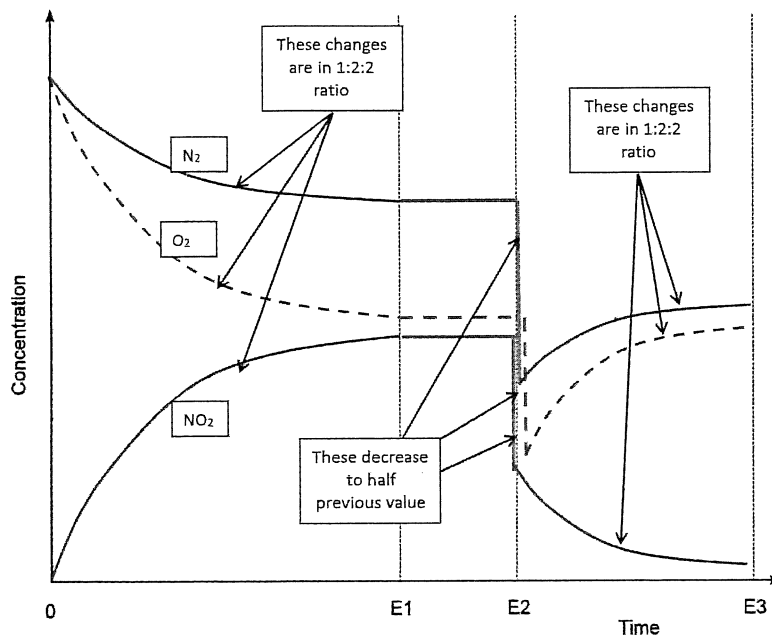
4.(2016:41) a) $K = \frac{p\text{NO}_2(\text{g})^2}{p\text{N}_2(\text{g})p\text{O}_2(\text{g})^2}$ (preferred) or $K = \frac{[\text{NO}_2]^2}{[\text{N}_2][\text{O}_2]^2}$

- b) No change in K

c) i) and ii)



d) and e)

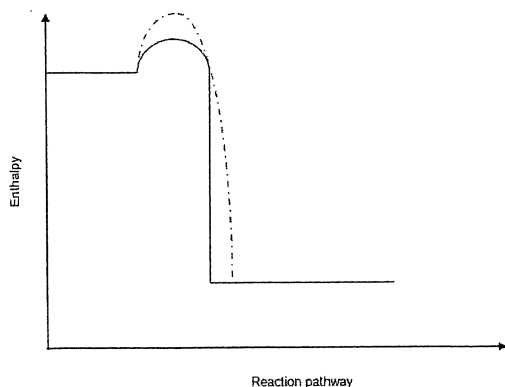


f) As the reactants are favoured, the equilibrium constant would be small. The activation energy would be large.
5.(2017:28) a)

Change imposed by the addition of	Effect on forward reaction rate when equilibrium	Effect on pH when equilibrium is re-established
a few drops of concentrated hydrochloric acid	increase	decrease
a few drops of concentrated lead(II) nitrate solution	decrease	decrease
distilled water	decrease	increase

b) The pH will decrease (fall). The increase in temperature provides more heat to the reaction. According to LCP the endothermic reaction will be favoured. This is the forward reaction. So there will be more products and the $[\text{H}_3\text{O}^+]$ will rise. Thus the pH will decrease.

6.(2017:33) a)



b) on diagram

c) The secret to this question is the phrase 'Typically, dynamite is between 25% to 50% nitroglycerin.' Given the molar masses are nearly identical we can see that for the same mass we will have the same mol. So dynamite gives 5 MJ and TNT gives 4MJ for the same number of mol. But the question asks for 'active ingredient' and in dynamite that is nitroglycerin at ... say 25%. So the nitroglycerine gives us 20MJ/mass. So the answer is nitroglycerine (not dynamite) produces 5 times as much energy as TNT.

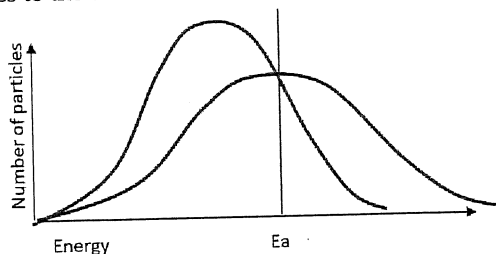
7.(2018:32) Because the question asked us to explain rather than predict we cannot use Le Châtelier's Principle and must use Collision Theory. (see the syllabus items 1.8 and 1.9 at the front of this book for clarification of this point)

If atmospheric carbon dioxide partial pressure increases then equation (1) will move to the right because there will be a higher frequency of successful collisions on the LHS and the forward rate will increase while the reverse rate will be unchanged.

This increase in aqueous carbon dioxide will increase the frequency of collisions and successful collisions on the LHS of equation (2) while the reverse rate is unchanged. The forward rate in equation (2) will therefore increase. This has the effect of using up carbonate ions.

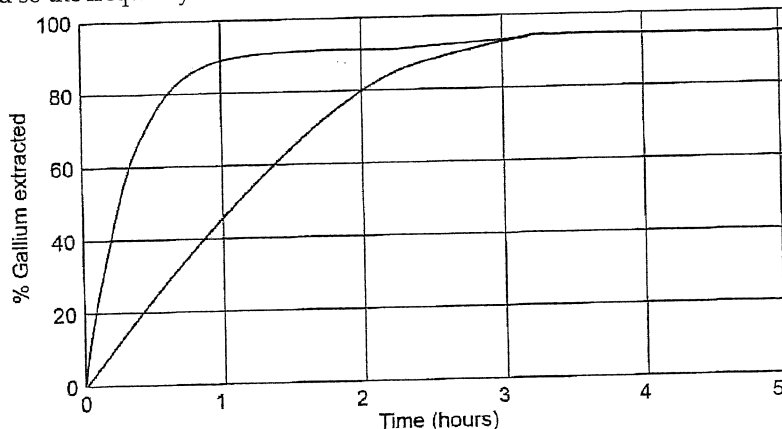
The decreased carbonate ion concentration will mean fewer collisions on the LHS of equation (3) and the forward rate will decrease while the backward rate remains unchanged. The effect will be the equation equilibrium moves to the left and dissolves some solid calcium carbonate and the rate of formation will decrease.

8.(2018:37) a)



As the temperature increases a larger population of particles have higher kinetic energy and they have energy that is higher or equal to the activation energy. Frequency of collisions increase as the particles are moving faster and so the frequency of successful collisions increases and the rate of reaction increases.

b)



c) Higher concentration of hydrochloric acid means there is a larger number of reactant particles per unit volume. Frequency of collisions increase, frequency of effective collisions increase hence it would take less time to complete the reaction and extract the gallium. Note there is NOT an increase in yield as the gallium is the limiting reagent. This is NOT an equilibrium reaction.

d) We want the gallium to be the limiting reagent and to extract all of it the acid must be INXS.

9.(2018:38) a) This question asks you to consider many factors. Best to answer using headings.

Temperature

Yield

- the reaction is exothermic and a low temperature would give a good yield

Rate

- high temperatures increase reaction rate for all reactions

Economics

- high temperatures are expensive – energy is expensive

Compromise

- We must choose a mid-temperature as a compromise between yield and rate and cost.

Pressure

Yield

- the reaction is a 5:2 mol ratio so increasing pressure would favour the products.

Rate

- high pressure increases the partial pressure of the gases and increases the reaction rate.

Economics

- High pressure is expensive in both energy and plant construction.

Compromise

- No compromise is needed for rate and yield but high pressure would be too expensive.

Catalyst

- A catalyst that is reusable, and increases rate will compensate for the lower temperature and lower pressure and yet encourage a high rate of reaction.

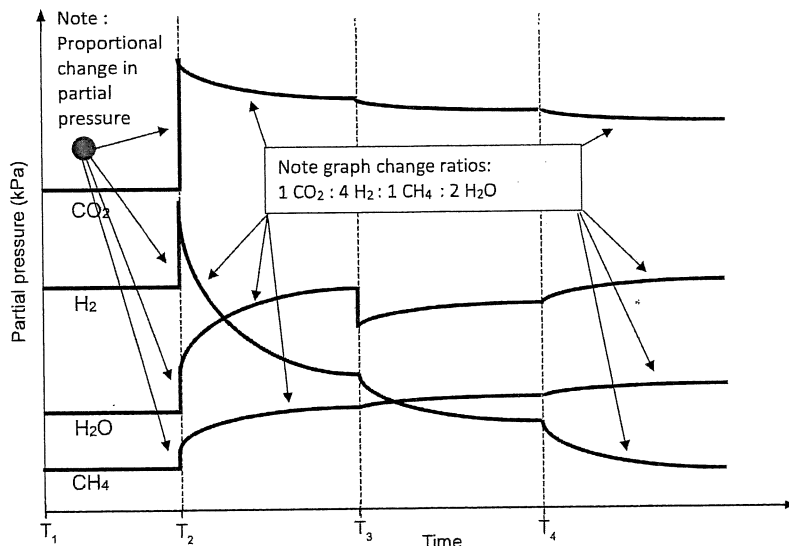
b)

Imposed change	Effect on methane yield (circle your answer)		
a suitable catalyst is added	increase	decrease	no effect
the volume of the reaction vessel is increased	increase	decrease	no effect
the temperature is increased	increase	decrease	no effect
methane is removed through a special valve as soon as it forms	increase	decrease	no effect
the partial pressure of carbon dioxide is decreased	increase	decrease	no effect

c) i) The system reached equilibrium at T_1 .

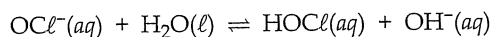
ii) At T_2 the volume of the system was decreased.

iii)



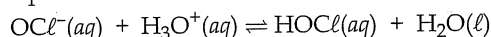
10.(2019:27) a) The word 'basic' means the pH is >7 - the explanation must therefore mention pH.

The OCl^- ion is the conjugate of the weak hypochlorous acid. The OCl^- ion undergoes hydrolysis, forming HOCl and $\text{OH}^-(\text{aq})$. This means the $[\text{OH}^-] > [\text{H}^+]$ and so the pH > 7 and so the solution is basic.

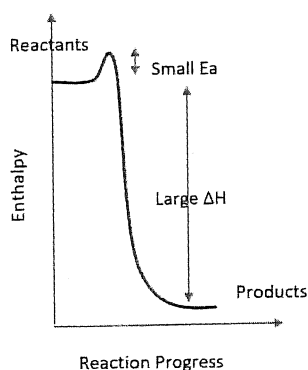


b) The addition of hydrochloric acid is not the reason the pool becomes more acidic. The reason is that the hydronium ions from the hydrochloric acid engage in the formation of $\text{HOCl}(\text{aq})$ from $\text{OCl}^-(\text{aq})$ and prevent it from undergoing hydrolysis and producing $\text{OH}^-(\text{aq})$.

When you complete an answer to a question, read the question again and then you see you need to state that the pH decreases and then write this equation.



c)



11.(2020:29) a) $K = \frac{[\text{H}_2]^4}{[\text{H}_2\text{S}]^2[\text{CH}_4]}$

b) Removing CH_4 reduces the partial pressure of CH_4 . There will be a decrease in the frequency of collisions between CH_4 and H_2S molecules so a decrease in frequency of successful collisions. The forward rate decreases. Initially the rate of the reverse reaction is not affected.

The rate of the reverse reaction, therefore, is greater than the rate of the forward reaction.

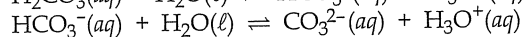
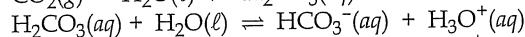
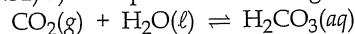
The equilibrium position, therefore, shifts to the left/equilibrium favours the reverse reaction.

c) Because 'predict' is used in the question we can use Le Châtelier's Principle. To answer this question satisfactorily you must refer to the graph and 'question (a)' in your answer. The graph indicates that as the temperature increases the number of moles of H_2 at equilibrium decreases. As temperature increases, an endothermic reaction is favoured and this is the reason the H_2 decreases. The equation could be rewritten as



Therefore $[H_2S]$ and $[CH_4]$ increase while $[H_2]$ decreases as shown in the formula in part (a), K decreases.

12.(2020:31) a) This question is dealing with ocean acidification



b) Again, reading the question closely is essential. We are asked to discuss shelled animals such as oysters, cockles etc. This question does not include corals as they do not have shells.

Organisms use shell for defence and $Ca^{2+}(aq)$ and $CO_3^{2-}(aq)$ are required for growth. With loss of $CO_3^{2-}(aq)$ by acidification shell is thinner and of less quality making them more vulnerable to predators.

c) (The Kyoto Protocol agreed to reduce global CO_2 emissions by 5.2% below 1990 levels by 2012. It was not achieved. We have increased emissions by 69% between 1990 and 2018.)

By abandoning the Kyoto agreement, increasing the partial pressure of CO_2 in the atmosphere, each of the equations above will be forced to the right-hand side. Thus, more CO_2 dissolves in the oceans, more carbonic acid forms, more hydrogen ions form.

13.(2020:34) a) Notice here we are asked whether we agree and we are not asked to state Le Châtelier's Principle. Also we are not asked to present both a heating and cooling discussion – that just wastes time as they are opposites. Also note this is a yield question – do not discuss collisions, rates or the compromise.

I agree. If the temperature is raised then more energy is available. The Haber equation produces energy (exothermic) so the reverse equation is endothermic and is favoured according to Le Châtelier's Principle to remove the excess heat. The result is that the yield of ammonia decreases.

b) This question is about rate and collision theory only – do not mention Le Châtelier's Principle, yield, the compromise nor comparative rates.

I agree. Higher temperatures mean more heat is available and the average kinetic energy of the particles increases.

Activation Energy More collisions have energy equal to or greater than the activation energy.

A greater *proportion* of the collisions are, therefore, successful.

Collisions Particles have more kinetic energy and are moving faster so collide more *frequently*.

There is a higher frequency of successful collisions. The reaction rate increases as the temperature increases.

Chapter 8: Acids and Bases

1.(2015:29) a) $c(H^+) = n/v = 8.50 \times 10^{-3} / 0.025 = 0.340 \text{ mol L}^{-1}$

$$pH = -\log[H^+] = -\log(0.340) = 0.469$$

$$b) n(OH^-) = cv = 0.3 \times 0.02 = 0.006 \text{ mol}$$

OH^- and H^+ react in 1:1 ratio

$$n(H^+) \text{ in excess} = 0.0085 - 0.006 = 2.5 \times 10^{-3} \text{ mol}$$

$$c(H^+) = 2.5 \times 10^{-3} / 0.045 = 0.0556 \text{ mol L}^{-1}$$

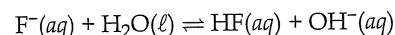
$$pH = -\log[H^+] = -\log(0.0556) = 1.26$$

2.(2015:40) a) i) Covalent bond ii) Hydrogen bond

iii) High electronegativity of F means the molecule is very polar – an extreme dipole-dipole bond.

b) The same pH means the same $[H^+]$. HF is a weak acid and does not ionise to the same extent as HCl. So a greater concentration of HF is needed to give the required $[H^+]$

c) Fluoride hydrolyses resulting in the formation of hydroxide ions producing a solution with a $pH > 7$



The chloride ion is the weak conjugate base of a strong acid (HCl) and is a neutral ion

d) i) Hydrolysis of propanoic acid:



When hydrogen ions are added to the buffer, the equilibrium will (by LCP) shift left to use up the added H^+ ions. So the overall change to $[H^+]$ is minimised and so the pH change is insignificant.

ii) Equal concentrations of acid / conjugate base and high concentrations of acid and conjugate base.

3.(2016:31) a)

Acidic	Neutral	Basic
NH_4Cl	$KCl, NaNO_3$	$KCN, Mg_3(PO_4)_2$